Theory Week 2 ~lschen

2	22/09/25	polar coord's, space curve description, linear momentum	1.1.3-1.2.1
	24/09/25	linear momentum balance, angular momentum balance	1.2.1-1.2.2
		exercise 2: single-particle kinematics & kinetics	

script sections

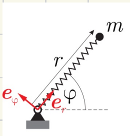
Orga

Mechanics 3 CAB G59

WhatsApp group

- Find all doc's on my website (link in WhatsApp bio)
- remember to do the STACK
- MIDTERM (week 8)
  - be in your group
  - counts for bonus (100 pts)
- Study Center
- recap videos from Dennis (i think they're released at end of the semester)

Polar coordinates

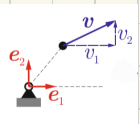


script p.8

Here basis are time-dependent

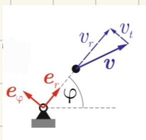
$\Rightarrow$ non-inertial frame $\Rightarrow \dot{e}_i \neq 0$

cartesian $\rightarrow$ polar



- $\boldsymbol{r}(t) = r(t)\cos\varphi(t)\boldsymbol{e}_1 + r(t)\sin\varphi(t)\boldsymbol{e}_2.$
- $\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = (\dot{r}\cos\varphi - r\dot{\varphi}\sin\varphi)\boldsymbol{e}_1 + (\dot{r}\sin\varphi + r\dot{\varphi}\cos\varphi)\boldsymbol{e}_2$
- $\boldsymbol{a} = (\ddot{r}\cos\varphi - 2\dot{r}\dot{\varphi}\sin\varphi - r\dot{\varphi}^2\cos\varphi - r\ddot{\varphi}\sin\varphi)\,\boldsymbol{e}_1 + (\ddot{r}\sin\varphi + 2\dot{r}\dot{\varphi}\cos\varphi - r\dot{\varphi}^2\sin\varphi + r\ddot{\varphi}\cos\varphi)\,\boldsymbol{e}_2.$

some proving (PS)



p.9

velocity and acceleration components in **polar coordinates** (r,φ):

$$\boldsymbol{v} = \dot{r}\boldsymbol{e}_r + r\dot{\varphi}\boldsymbol{e}_\varphi, \qquad \boldsymbol{a} = (\ddot{r} - r\dot{\varphi}^2)\boldsymbol{e}_r + (2\dot{r}\dot{\varphi} + r\ddot{\varphi})\boldsymbol{e}_\varphi$$

$\underline{v} = \begin{pmatrix} \dot{r} \\ r\dot{\varphi} \end{pmatrix}$ $\quad a = \begin{pmatrix} \ddot{r} - r\dot{\varphi}^2 \\ 2\dot{r}\dot{\varphi} + r\ddot{\varphi} \end{pmatrix}$

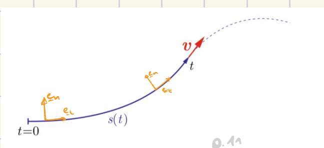
remember:

$\omega = \dot{\varphi}$ = ang. vel.

$\dot{\omega} = \ddot{\varphi}$ = ang. acc.

Space curves

- general description of 3D particle trajectory



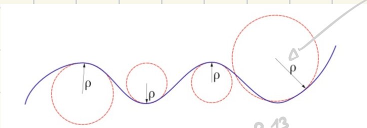
p.11

$$\boldsymbol{v} = \dot{s}\boldsymbol{e}_t, \qquad \boldsymbol{a} = \ddot{s}\boldsymbol{e}_t + \frac{v^2}{\rho}\boldsymbol{e}_n \qquad \text{with} \qquad \boldsymbol{e}_t = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s} = \frac{\boldsymbol{v}}{v}, \quad \boldsymbol{e}_n = \frac{\dot{\boldsymbol{e}}_t}{|\dot{\boldsymbol{e}}_t|} = \frac{\rho}{v}\dot{\boldsymbol{e}}_t$$

unit tangent vector

centripetal acc.

curvature



p.13

1.2 Kinetics

Def. kinetics: How single particle motion depends on forces acting on it

Newton's Axioms

$\underline{P} = m \cdot \underline{v}$ // $\underline{F} = m \cdot \underline{a}$

1st axiom

absence of forces $\Rightarrow$ linear momentum conserved

$\boxed{\sum_i \underline{F}_i = 0} \Rightarrow \frac{d}{dt}(m \cdot \underline{v}) = 0 \Rightarrow \boxed{m\underline{v} = \text{const}}$

2nd axiom

rate of change of lin. mom. = applied (net) force

same same

balance of linear momentum for a particle of mass m:

$$\sum_i \boldsymbol{F}_i = \dot{\boldsymbol{P}} = \frac{\mathrm{d}}{\mathrm{d}t}(m\boldsymbol{v}) \overset{m=\text{const.}}{\Rightarrow} \sum_i \boldsymbol{F}_i = m\boldsymbol{a}$$

$\Rightarrow \boxed{\text{LMB}}$

think vacuum vs gravity

$\Rightarrow$ to change lin. mom., force is needed !!!

↳ 3 important scenarios:
- $\underline{v} = 0 \Rightarrow \underline{F} = 0$ (static)
- mass = const $\Rightarrow \underline{F} = m \cdot \underline{a}$ (otherwise rocket eq.)
- $\sum \underline{F} = 0 \Rightarrow \underline{P} = m \cdot \underline{v}$ = const (1st axiom)

3rd axiom

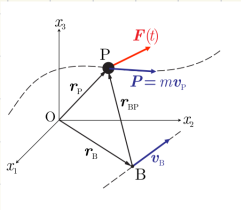
$\boxed{\text{Actio = Reactio}}$ ~ Mazza

exp. 1.8 (sliding block p. 17) $\Longrightarrow$

kinetic friction:

$$\boldsymbol{R} = -\mu|\boldsymbol{N}|\frac{\boldsymbol{v}}{|\boldsymbol{v}|}, \qquad |\boldsymbol{R}| = \mu|\boldsymbol{N}|$$

Balance of angular momentum (AMB)



balance of angular momentum of a particle with respect to point B:

$$M_B = \dot{H}_B + v_B \times P \qquad \text{with} \qquad H_B = r_{BP} \times P, \quad P = mv_P$$

resultant torque

$\underline{M}_B = \underline{r}_{BP} \times \sum_i \underline{F}_i$

same same

ang. mom.

w.r.t. point B

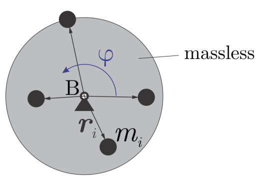
OR as seen before $F=0 \Rightarrow M=0$

important scenarios:

- $v = a = 0$ (static) $\rightarrow \dot{H}_B = 0,\ P = 0 \Rightarrow \underline{M}_B = 0$
- B fixed ($\underline{v}_B = 0$) / $v_B \parallel v_P$ (B&P move in parallel)
  $\hookrightarrow v_B \times P = 0 \Rightarrow M_B = \dot{H}_B$
- $\underline{M}_B^{ext} = 0$ & $v_B = 0$ / $v_B \parallel v_P \rightarrow \dot{H}_B = 0 \Rightarrow r_{BP} \cdot v_P = \text{const}$
- $m = 0$ (massless body) $\rightarrow P = 0 \rightarrow H_B = 0 \rightarrow \dot{H}_B = 0$
  $\Rightarrow \underbrace{\sum_i \underline{F}_i = 0}_{\text{from LMB}},\ \underline{M}_B = 0$

$\hookrightarrow$ always good to chose B to have these special cases! (results in easy math)

Ex. 2.3



(script p.53)

Particle on rot. disc

special case of a **rotation in 2D** around a fixed point B at a distance R:

$$M_B = I_B \ddot{\varphi} \qquad \text{with} \qquad I_B = mR^2$$

When do I use what? p.22 script

AMB mostly for rotational problems with known forces (often useful to use polar coordinates)

remember LMB & AMB describe the same thing

many examples in script, which may be helpful

LMB p.50

AMB p.53-56